ZIMBABWE SCHOOL EXAMINATIONS COUNCIL (style)

ACADEX PRACTICE EXAMINATION
A-LEVEL · PURE MATHEMATICS

MATHEMATICS 6042/2

PAPER 2 November 2024
3 hours
PAPER 2 — STRUCTURED QUESTIONS — CALCULATOR ALLOWED

These are **original ACADEX questions** written to match ZIMSEC syllabus structure, mark allocations and command words. They are **not** leaked or copied official papers. This script is unique to November 2024 6042/2.

Additional materials: A calculator may be used unless a question forbids it.
INSTRUCTIONS: Answer all questions. Full marks are not given for answers only — show method.

1. Find the stationary points of $y = x^3 - 6x^2 + 9x + 1$ and determine their nature. **[10]**

2. Find ∫ x e^x dx using integration by parts. **[10]**

3. Solve $dy/dx = 3y$ given that $y = 2$ when $x = 0$. **[10]**

4. $x = 2t$, $y = t^2$. Find dy/dx in terms of t, and the equation of the tangent at $t = 1$. **[10]**

5. A(1,0,2), B(3,4,2). Find AB and |AB|. Show AB is perpendicular to (2, −1, 0) or not. **[10]**

6. Show that $f(x)=x^3 - x - 1$ has a root in [1, 2]. Take one Newton–Raphson step from x0=1. **[8]**

7. Solve $|2x - 3| < 5$. **[8]**

8. Solve $2 \sin \theta = 1$ for $0 \le \theta \le 360°$. **[8]**

9. A GP has first term 5 and common ratio 1/2. Find S_∞ and the smallest n with S_n > 9.5. **[8]**

10. $x^2 + y^2 = 25$. Find dy/dx at (3, 4). **[8]**

11. Find the area between $y = 4x - x^2$ and the x-axis from x=0 to x=4. **[8]**

12. For small θ in radians, write approximations for $\sin\theta$, $\cos\theta$, $\tan\theta$. Estimate $\sin 0.1$. **[8]**

END OF QUESTION PAPER

Worked solutions and mark-scheme notes are inside the ACADEX app (Extract & Study).